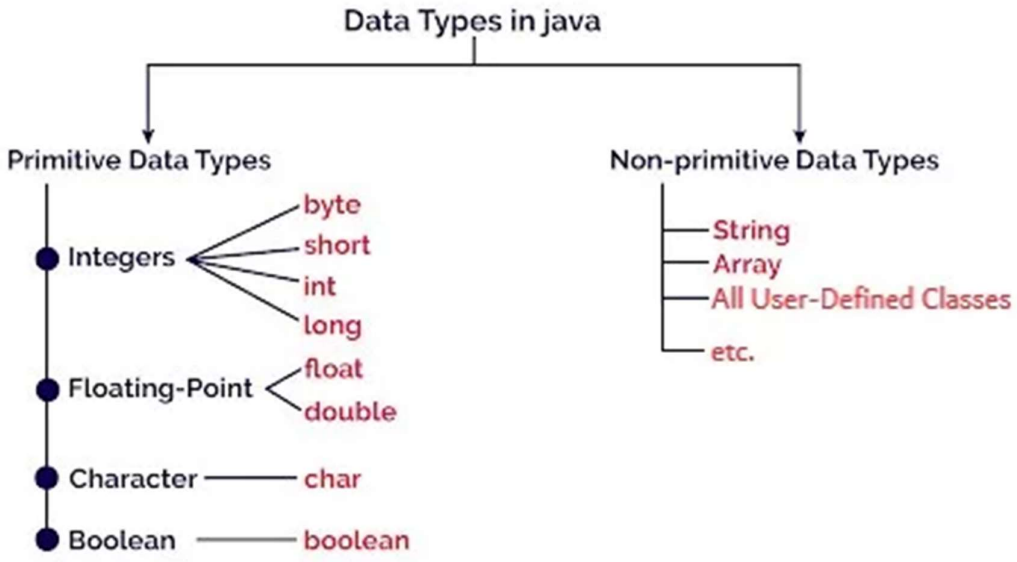
DATATYPE IN JAVA



Primitive Data Type :-

- A primitive data type specifies the **type of a variable** and the **kind of values it can hold.**

S.No	Data Type	Description/Nature	Storage Value	Size
1.	Byte	Arithmetic Type	Integer	1 Byte
2.	Short	Arithmetic Type	Integer	2 Byte
3.	Int	Arithmetic Type	Integer	4 Byte
4.	Long	Arithmetic Type	Integer	8 Byte
5.	Float	Arithmetic Type	Float/Decimal	4 Byte
6.	Double	Arithmetic Type	Float/Decimal	8 Byte
7.	Boolean	Boolean Type(0/1)	True or False	1 Bit
8.	Char	Character Type	Character	2 Byte

Here are the eight primitive data types in Java:

- **byte:-** An **8-bit** signed **two's complement integer**. Its range is from **-128 to 127.**
- **short:-** A **16-bit** signed **two's complement integer**. Its range is from **-32,768 to 32,767.**
- **int:-** A **32-bit** signed **two's complement integer**. This is the default integer type and is commonly used for general-purpose integer values.
- **long:-** A **64-bit** signed **two's complement integer**. Used when int is not large enough to hold the desired value.
- **float:-** A **32-bit** is **floating-point** number.
- **double:-** A **64-bit** is **floating-point** number. This is the default floating-point type and offers higher precision than float.
- **char:-** A **16-bit** Unicode character. Used to store **single characters.**
- **boolean:-** Represents a **logical value**, which can be either **true or false.**

VARIABLE IN JAVA

What is Variable :-

- It is a container where **value will be store.**
- Memory location for data value.
- Name of Memory Location.

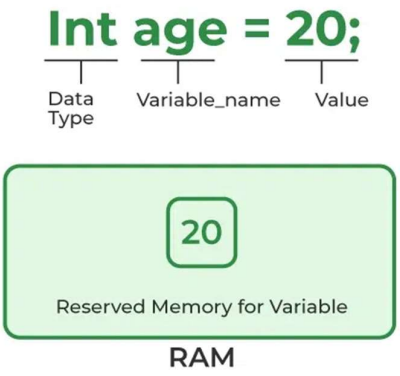
Rules of Variable :-

- It starts with **only character** and **underscore**. For ex. **a** and **_a**.
- It can't start with **number**, **special character**, and **keyword**. For ex. **int a=1**
- It can be **Alphanumeric.**
- It doesn't contain **any spaces.**

Syntax and Example :-

Syntax :- Datatype VariableName = value:
Example :- Byte age = 23;

Syntax :- Datatype VariableName = value:
Example :- Boolean b = True;



Implementation

Input :-

```java
package basicconcepts;

public class datatypes {

    public static void main(String[]args)
    {
        System.out.println("datatypes");

        byte num=127;
        System.out.println("byte value is: "+num);

        short a=12345;
        System.out.println("short value is:"+a);

        int b=8282828;
        System.out.println("int value is:"+b);

        long c=82828282828282828281;
        System.out.println("long value is:"+c);

        float f=857.7f;
        System.out.println("float value is: "+f);

        double d=7567454;
        System.out.println("double value is: "+d);

        boolean result=true;
        System.out.println("Boolean Value is: "+result);

        char ch='p';
        System.out.println("Character value is: "+ch);

        String s="Nikhil Shinde";
        System.out.println("String is: "+s);
    }
}
```

Output :-

```
datatypes
byte value is: 127
short value is:12345
int value is:8282828
long value is:8282828282828282828
float value is: 857.7
double value is: 7567454.0
Boolean Value is: true
Character value is: p
String is: Nikhil Shinde
```